

Unit – 5 : Session – 2 : SLO – 1 | Constructing DFA and NFA for Basic Languages**I. Questions & Answers**

1. DFA for strings over $\{0,1\}$ ending with '0' — states $\{q_0, q_1\}$; start = q_0 , accepting = $\{q_1\}$. Transitions: $q_0 \xrightarrow{0} q_1$, $q_0 \xrightarrow{1} q_0$, $q_1 \xrightarrow{0} q_1$, $q_1 \xrightarrow{1} q_0$. The machine is in q_1 exactly when the last symbol read was 0.
2. NFA for strings starting with '1' — states $\{q_0, q_1\}$; start = q_0 , accepting = $\{q_1\}$. Transitions: $q_0 \xrightarrow{1} q_1$, and q_1 loops on 0 and 1. (From q_0 , input 0 has no transition — the string is rejected.)
3. Can an NFA have ϵ -transitions? — Yes. An ϵ -NFA may change state without consuming any input symbol; ϵ -transitions are the key tool in Thompson's construction, and every ϵ -NFA can be converted to an equivalent DFA.